



Thermodynamic modeling of the V–Zr system supported by key experiments



Jiaxin Cui, Cuiping Guo, Lei Zou, Changrong Li, Zhenmin Du*

Department of Materials Science and Engineering, University of Science and Technology Beijing, Beijing 100083, PR China

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ABSTRACT

The V–Zr system was investigated by experiments and thermodynamic modeling. In experiments, 10 crucial alloys were selected, and the microstructure, crystal structure and phase transformation temperatures of the alloys were obtained using scanning electron microscopy (SEM) with energy dispersive spectrometer (EDS), X-ray powder diffraction (XRD) and differential thermal analysis (DTA). A thermodynamic modeling of the V–Zr system was then performed by considering the reviewed literature data and incorporating the present experimental results. The solution phases, liquid, hcp (Zr) and bcc(V, Zr), were modeled with the substitutional solution model, and the compound V_2Zr was treated as $(V, Zr)_2(V, Zr)$ by means of two-sublattice model. A set of self-consistent thermodynamic parameters of the V–Zr system was obtained.

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1. Introduction

V-based alloys with body centered cubic (bcc) structure and Zr-based alloys with AB_2 type Laves phase have been extensively investigated due to their remarkable hydrogen storage capacity, easy activation and moderate working conditions [1–4]. As a kind of novel hydrogen storage alloy, the Laves phase related to bcc solid solution alloy has received more attention in recent years due to its high hydrogen storage capacity of more than 2.0 wt% and fast hydrogen absorption and desorption kinetics at ambient temperature and pressure [4,5]. The V–Zr multiphase alloys commonly consist of C15 Laves phase and bcc solution phase. The synergetic effects of these phases could improve the performance of their hydrogen storage properties [6]. So it is very important to study the V–Zr phase diagram for the research of the hydrogen storage alloys.

The present investigation is to provide new phase diagram data for the V–Zr system on the basis of the reliable thermodynamic description of this binary system. The experimental investigation of the V–Zr system was measured by SEM-EDS [7], DTA [8] and XRD [9]. Additionally, the V–Zr was modeled by means of the CALPHAD technique on the present experimental results and the available literature reports [10–16].

2. Literature review

2.1. Phase diagram data

The phase diagram of the V–Zr system was investigated experimentally by many investigators [10–12] and assessed by Masalski and Okamoto [13]. This system was characterized by a non-stoichiometric compound V_2Zr which melts incongruently by a peritectic reaction $V_2Zr \rightarrow \text{liquid} + \text{bcc}(V)$. V_2Zr was widely studied on its structure and physical properties [14–17]. Naskidashvili et al. [14] studied the thermodynamic properties of V_2Zr by differential calorimetric measurements in the region of 80–300 K. However, the authors of Ref. [14] themselves pointed out that their results were not precise absolute values of heat capacity. There were three invariant reactions in the V–Zr system, a peritectic reaction, $\text{liquid} + \text{bcc}(V) \rightarrow V_2Zr$, a eutectic reaction, $\text{liquid} \rightarrow V_2Zr + \text{bcc}(Zr)$, and a eutectoid reaction, $\text{bcc}(Zr) \rightarrow V_2Zr + \text{hcp}(Zr)$. Neither of the peritectic point nor the eutectic point was determined by earlier trials [10–12].

2.2. Thermodynamic optimization results in literature

Thermodynamic optimization results [18–20] of the V–Zr system showed a big difference due to the lack of experimental data. The V–Zr system was modeled firstly by Korb et al. [18]. However, the V_2Zr phase was treated as a stoichiometric compound and the differences between the calculated composition of liquid in the invariant reactions and the experimental data were evident. Later on, the V–Zr system was optimized by Servant [19] and Zhao et al. [20]. At high temperature, both V and Zr are of bcc structure, so the

* Corresponding author.

E-mail address: zmdu2@hotmail.com (Z. Du).

curve of bcc phase should be closed at high temperature. But the calculated bcc phase had an inverted artificial miscibility gap at high temperature according to thermodynamic parameters of Servant [19] and Zhao et al. [20].

3. Experimental procedures

The phase equilibria of the V–Zr system were revised with 10 samples by SEM-EDS [7], DTA [8] and XRD [9]. The compositions of the tested alloys were detailed in Table 1. To ensure accuracy, the compositions of ten alloys were all reconfirmed by EDS of SEM-EDS method [7]. The purity of Titanium, Vanadium and Zirconium was 99.995, 99.95 and 99.96 wt%, respectively. All of samples were prepared by arc melting on a water-cooled copper hearth under an argon atmosphere. To ensure homogeneity, the ingots were melted no less than four times and inverted after every melting. Samples with melting losses not exceeding 0.5 wt% were chosen for the experiments.

The samples for the binary V–Zr system were cut into two halves using wire-electrode. One half was used for determining microstructure and phase composition by SEM with EDS [7] and crystal structure by XRD [9]. The SEM measurement was carried out on a 20 kV using a ZEISS EVO 18 Special Edition microscope. The composition of each phase was measured by Bruker 5010 EDS for SEM. The precision was 1% when the content of the component in the tested phase was more than 0.1%. The XRD was used to identify the crystal structures of the constituent phases. The XRD measurement was carried out on a Rigaku rotatory anode harrow X-pert diffractometer using Cu K α radiation at 40.0 kV and 150 mA. The data was collected in the range of 2θ from 20° to 100° with a step size of 0.02°. The other half of samples was cut into

flakelets for determining phase transition temperature by DTA [8], which was performed on a SDT Q600 (TA instruments). Zirconia crucibles instead of Alumina crucibles were used for presenting samples to reacting with crucibles. In order to obtain the better thermal conduction and thermal diffusion, the samples were cut into slices that just covered the bottom of the crucible and the heating rate was as slow as 10 °C/min. When different heating rates were tried on trial and error, the peak of 1070 K was too flat to be observed at the heating rate less than 5 °C/min, and the peaks of the DTA curve at the heating rate of 15 °C/min was almost as the same as those of 10 °C/min, but the temperature precision at the heating rate of 10 °C/min was more accurate with less over-heat. Thus, the measurements were performed between room temperature to 1350 °C with the heating rate of 10 °C/min under an argon atmosphere. The weight of the DTA samples are shown in Table 1. The invariant reaction temperatures were determined from the intersection of the tangent with the onset of the thermal effects and the valley bottom during the heating step.

4. Thermodynamic model

4.1. Pure elements

The Gibbs energy function $G_i^\phi(T) = {}^0G_i^\phi(T) - H_i^{\text{SER}}(298.15 \text{ K})$ for the pure element i ($i = \text{V}$ and Zr) in the ϕ ($\phi = \text{bcc}$ and hcp) phase is described by the following equation,

$$G_i^\phi(T) = a + bT + cT \ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where $H_i^{\text{SER}}(298.15 \text{ K})$ is the molar enthalpy of the element i at 298.15 K in its stable element reference (SER) state, bcc for V, and

Table 1
Compositions and solidification path of the V–Zr alloys.

No.	Alloy composition (at%)		Phases	Phase composition (at%)		Solidification path	Weight of DTA samples (mg)	DTA
	V	Zr		V	Zr			
#1	62	38	bcc(V)	97.01	2.99	liquid $\rightarrow$ bcc(V) liquid + bcc(V) $\rightarrow$ V ₂ Zr liquid $\rightarrow$ V ₂ Zr + bcc(Zr) bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)	14.65	– 1585 1537 –
				97.17	2.73			
				97.05	2.95			
			V ₂ Zr	64.56	35.44			
				67.17	32.83			
				66.10	33.90			
#2	57	43	V ₂ Zr	68.20	31.80	liquid $\rightarrow$ V ₂ Zr	15.10	–
				68.25	31.75	liquid $\rightarrow$ V ₂ Zr + bcc(Zr)		1537
				66.85	33.15	bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		–
#3	55	45	V ₂ Zr	67.08	32.92	liquid $\rightarrow$ V ₂ Zr	13.89	1586
				65.90	34.10	liquid $\rightarrow$ V ₂ Zr + bcc(Zr)		1537
				65.68	34.32	bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		–
#4	54	46	V ₂ Zr	65.33	34.67	liquid $\rightarrow$ V ₂ Zr	15.25	1584
				65.97	34.03	liquid $\rightarrow$ V ₂ Zr + bcc(Zr)		1535
				67.98	32.02	bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		–
#5	52	48	V ₂ Zr	65.49	34.51	liquid $\rightarrow$ V ₂ Zr	11.62	–
				67.24	32.76	liquid $\rightarrow$ V ₂ Zr + bcc(Zr)		1537
				67.15	32.85	bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		–
#6	50	50	V ₂ Zr	66.50	33.50	liquid $\rightarrow$ V ₂ Zr	14.13	–
				64.61	35.39	liquid $\rightarrow$ V ₂ Zr + bcc(Zr)		1538
				64.73	35.27	bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		–
#7	46	54				liquid $\rightarrow$ V ₂ Zr + bcc(Zr)	14.98	1538
						bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		1070
						liquid $\rightarrow$ bcc(Zr)		–
#8	45	55				liquid $\rightarrow$ V ₂ Zr + bcc(Zr)	13.74	1538
						bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		1070
						liquid $\rightarrow$ bcc(Zr)		–
#9	38	62	bcc(Zr)	7.42	92.58	bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)	13.60	1070
				7.55	92.44	liquid $\rightarrow$ bcc(Zr)		–
				7.36	92.64	liquid $\rightarrow$ V ₂ Zr + bcc(Zr)		1538
#10	32	68	bcc(Zr)	9.60	90.40	bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)	12.46	1070
				9.31	90.69	liquid $\rightarrow$ bcc(Zr)		–
				9.39	90.61	liquid $\rightarrow$ V ₂ Zr + bcc(Zr)		1538
						bcc(Zr) $\rightarrow$ V ₂ Zr + hcp(Zr)		1070

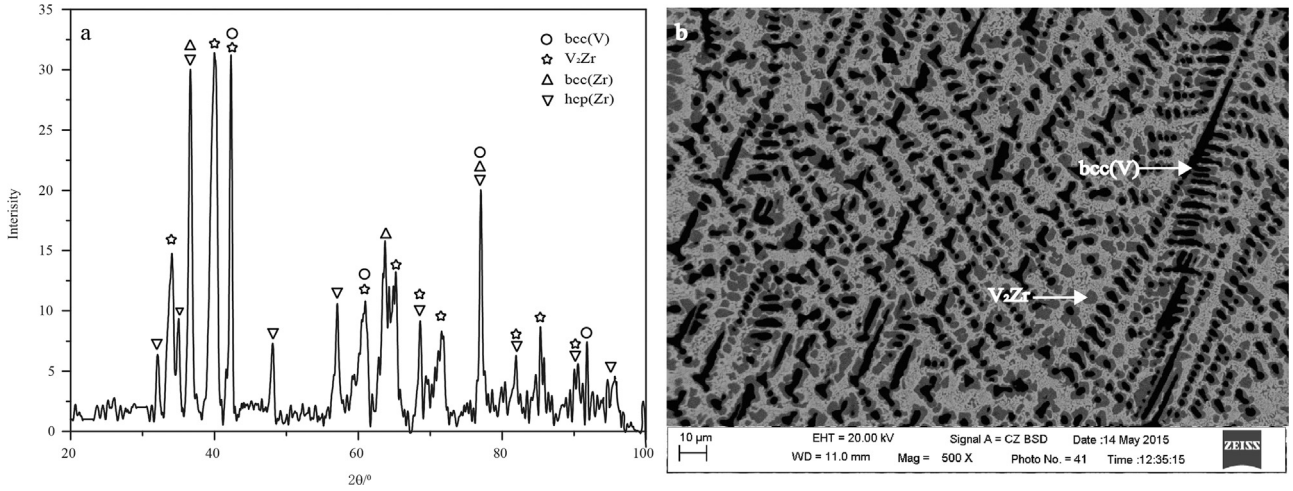


Fig. 1. (a) XRD diffraction pattern of alloy #1 (V-38 at% Zr) and (b) BEI micrograph of alloy #1 (V-38 at% Zr).

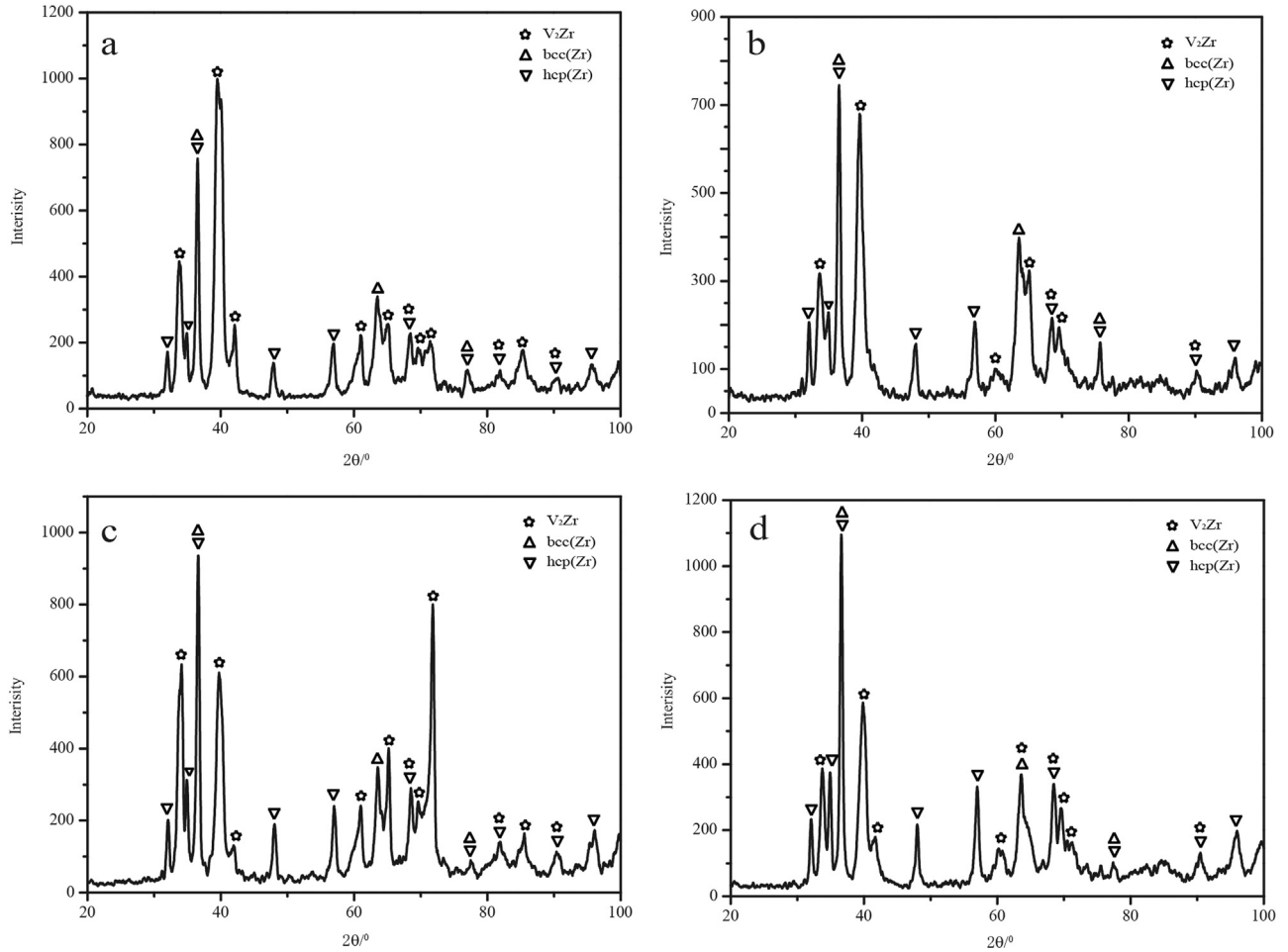


Fig. 2. XRD diffraction pattern of alloys in the V-Zr system (a) alloy #2 (V-43 at% Zr) (b) alloy #3 (V-45 at% Zr) (c) alloy #5 (V-48 at% Zr) and (d) alloy #6 (V-50 at% Zr).

hcp for Zr. The Gibbs energy of the element i ($i=V$ and Zr), $G_i^\phi(T)$, in standard element reference (SER) state, is denoted by $GHSER_i$, i.e.,

$$GHSER_i = {}^0G_i^\phi(T) - H_i^{SER}(298.15\text{ K}) \quad (2)$$

In the present work, the Gibbs energy functions for the elements V and Zr are taken from the SGTE (Scientific Group Thermochemical data Europe) database of pure elements [21].

4.2. Solution phases

In the V-Zr system, there are three solution phases: liquid, bcc and hcp. Their molar Gibbs energies are described by the following expression:

$$G_m^\phi(T) = x_V G_V^\phi(T) + x_{Zr} G_{Zr}^\phi(T) + RT(x_V \ln x_V + x_{Zr} \ln x_{Zr}) + {}^E G_m^\phi \quad (3)$$

where, R is the gas constant, x_V and x_{Zr} are the mole fractions of

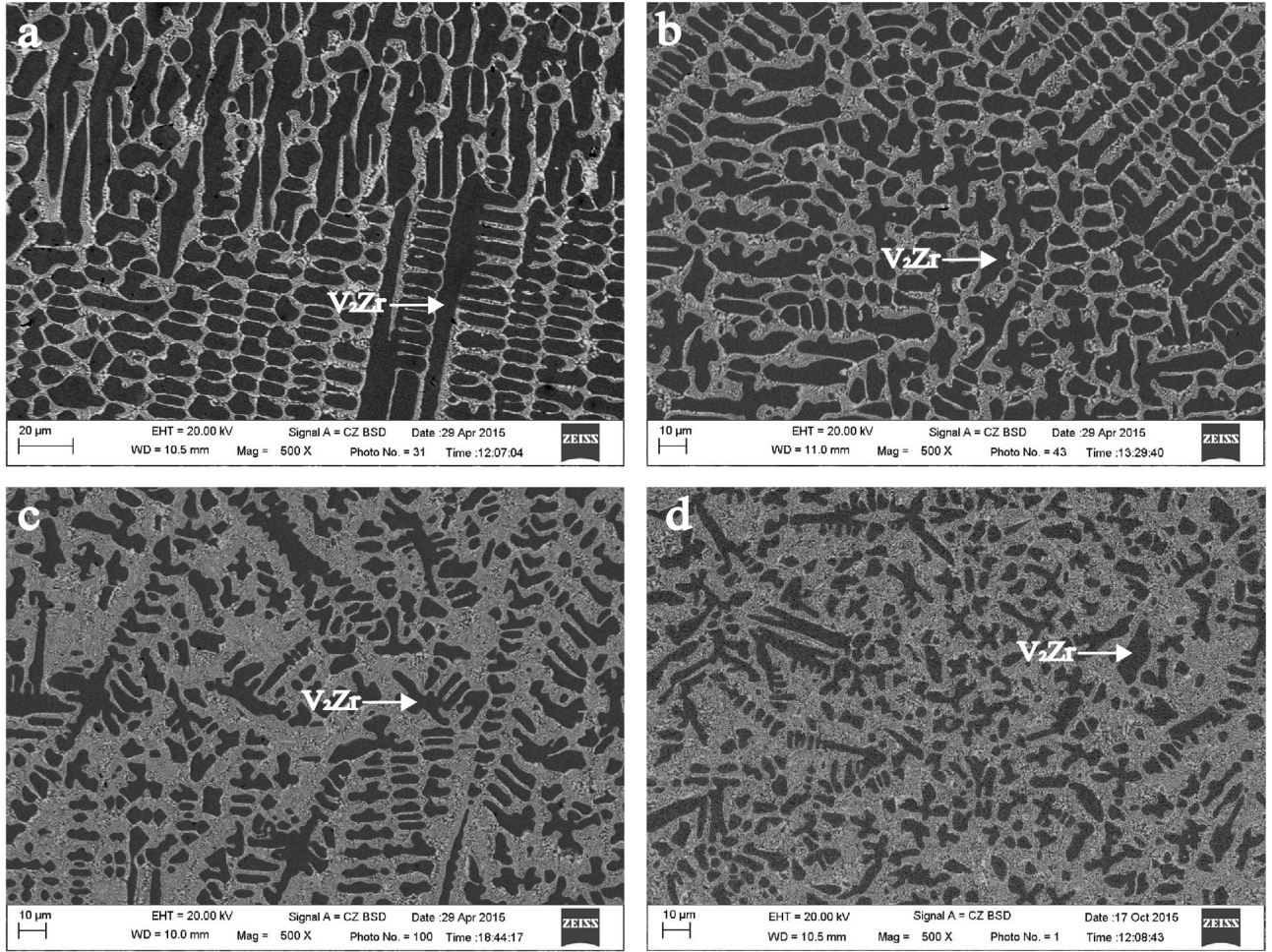


Fig. 3. BEI micrograph of alloys in the V–Zr system (a) alloy #2 (V–44 at% Zr) (b) alloy #3 (V–46 at% Zr) (c) alloy #5 (V–50 at% Zr) and (d) alloy #6 (V–50 at% Zr).

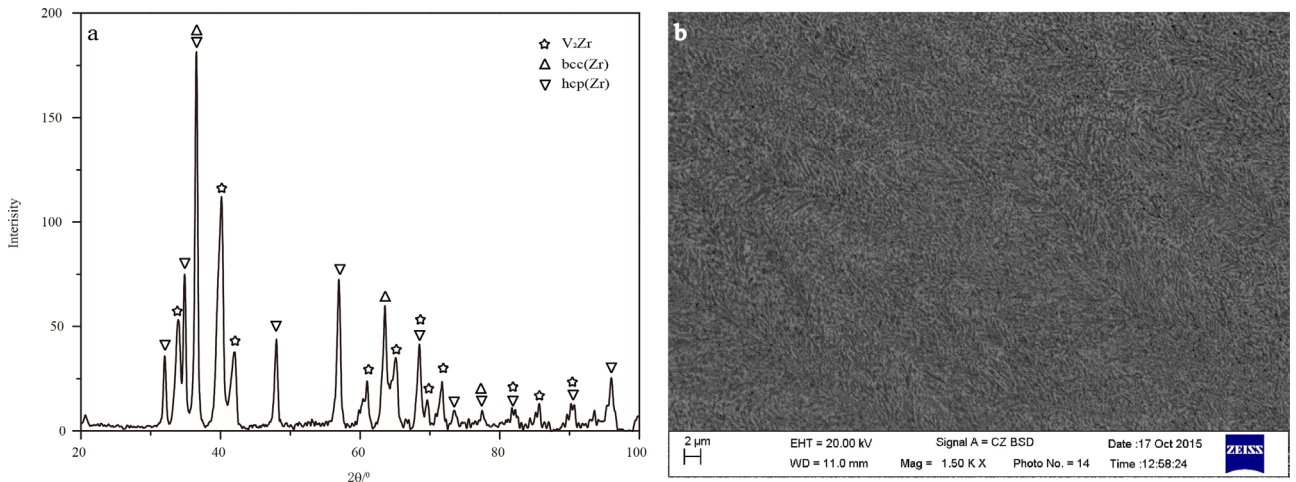


Fig. 4. (a) XRD diffraction pattern of alloy #7 (V–54 at% Zr) and (b) BEI micrograph of alloy #7 (V–54 at% Zr).

the elements V and Zr, respectively. E_m^ϕ is the excess Gibbs energy, which can be expressed by the Redlich–Kister polynomial,

$$E_m^\phi = x_V x_{Zr} \sum_j {}^j L_\phi (x_V - x_{Zr})^j \quad (4)$$

where ${}^j L_\phi$ is the interaction parameter between elements V and Zr. Its general form is,

$$L^\phi = a + bT + cT \ln T + dT^2 + eT^3 + fT^{-1} \quad (5)$$

where a, b, c, d, e and f are the parameters to be optimized. But in most case only the first one or two terms are used according to the temperature dependence of the experimental data. a and b are optimized in this work.

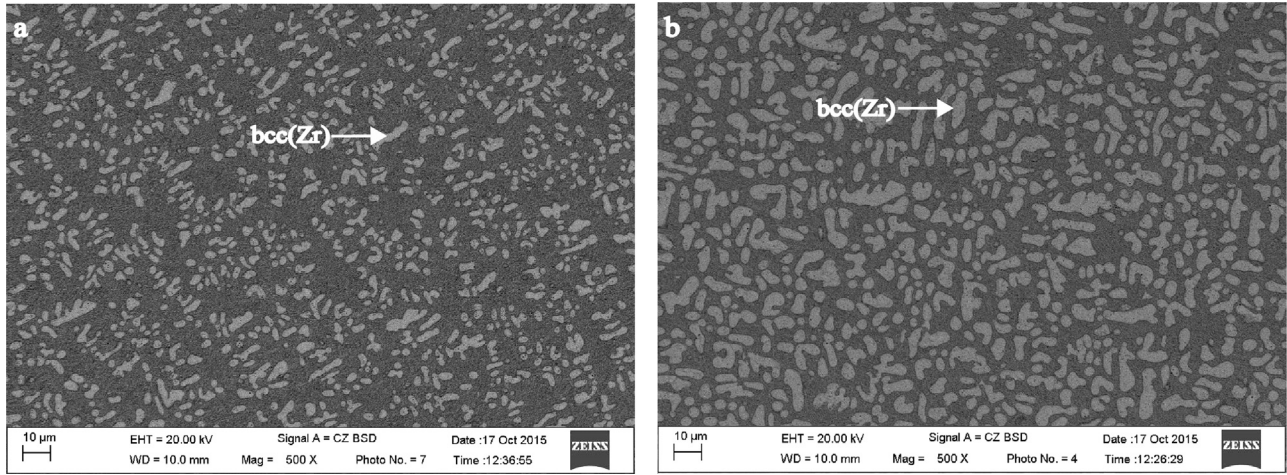


Fig. 5. BEI micrograph of alloys in the V–Zr system (a) alloy #9 (V–62 at% Zr) and (b) alloy #10 (V–68 at% Zr).

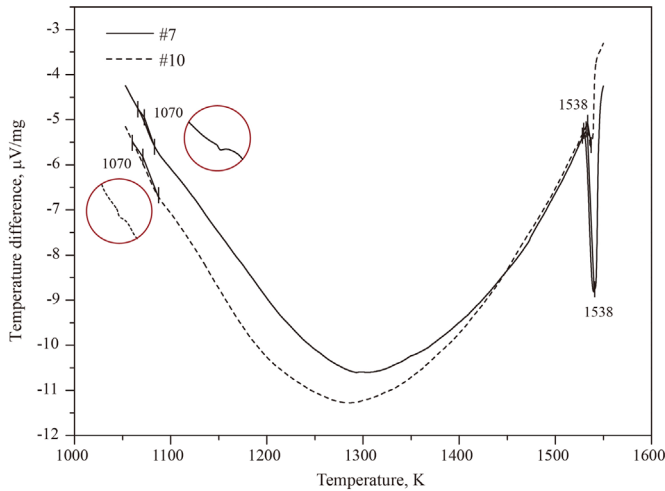


Fig. 6. DTA heating curves of alloy #3 (V–46 at% Zr) and alloy #4 (V–48 at% Zr) in the V–Zr system.

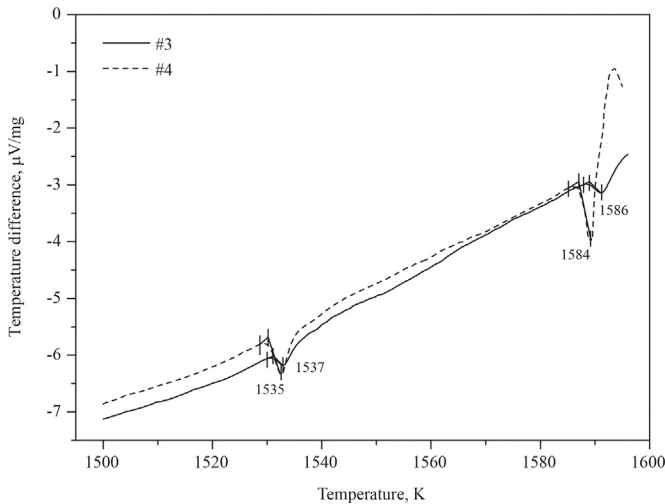


Fig. 7. DTA heating curves of alloy #7 (V–54 at% Zr) and alloy #10 (V–68 at% Zr) in the V–Zr system.

4.3. Intermetallic compound V_2Zr

The compound V_2Zr in the V–Zr system has a homogeneity range and is treated as $(V, Zr)_2(V, Zr)$ by a two-sublattice model

[22,23]. The Gibbs energy per mole of formula is expressed as follows,

$$G_m^{V_2Zr} = y'_V y'_Z G_{V:V}^{V_2Zr} + y'_V y'_Z G_{V:Zr}^{V_2Zr} + y'_Z y'_V G_{Zr:V}^{V_2Zr} + y'_Z y'_V G_{Zr:Zr}^{V_2Zr} + 2RT(y_V \ln y_V + y_Z \ln y_Z) + RT(y_V \ln y_V + y_Z \ln y_Z) + y'_V y'_Z \left[y'_V \sum_j j_L^{V_2Zr} (y'_V - y'_Z)^j + y'_Z \sum_j j_L^{V_2Zr} (y'_V - y'_Z)^j \right] + y'_V y'_Z \left[y'_V \sum_j j_L^{V_2Zr} (y'_V - y'_Z)^j + y'_Z \sum_j j_L^{V_2Zr} (y'_V - y'_Z)^j \right] \quad (6)$$

where y'_* and y'_* are the site fraction of each species (V, Zr) on the first and second sublattices, respectively; $G_{*:~*}^{V_2Zr}$ represents the Gibbs energy of the constituent compound of V_2Zr when the first and the second sublattices are occupied by only one element V or Zr, respectively; $j_L^{V_2Zr}$ and $j_L^{V_2Zr}$ present the j th interaction parameters of V_2Zr on the first and second sublattices, respectively.

5. Results and discussion

5.1. Experimental investigation

The phase transformation temperatures in all of V–Zr alloys are investigated by DTA in the present work, and the results are summarized in Table 1.

Fig. 1(a) shows the diffraction pattern of alloy #1 (V–38 at% Zr), and the diffraction peaks of bcc(V), V_2Zr , bcc(Zr) and hcp(Zr) are observed. Fig. 1(b) shows BEI micrograph of alloy #1. Three different microstructure regions the dark dendritic microstructure, the gray phase attached to the dark dendrite and the mixture of the gray and the bright phases, are observed, showing both the typical peritectic and the typical eutectic characteristics. In the microstructure, the dark dendrite microstructure is the primary solidification phase with a composition of V–2.99 at% Zr by EDS and the bcc(V) structure. The gray phase is V_2Zr with the composition of V–35.44 at% Zr by EDS.

For the alloys #2–#6, the primary solidification phase is identified to be V_2Zr as shown in Fig. 2 and Fig. 3. Fig. 2 shows the diffraction patterns of alloys #2–#6, and the diffraction peaks of V_2Zr , bcc(Zr) and hcp(Zr) phases are observed. Fig. 3(a) presents BEI micrograph of alloy #2 (V–43 at% Zr). Two different microstructure regions are observed. The dark dendrite microstructure is the primary solidification phase, and its composition measured by EDS is V–31.80 at% Zr, and is that of the V_2Zr phase. However, bcc(Zr) and hcp(Zr) phases can not be differentiated in Fig. 3. The

Table 2

Calculated invariant reactions in the V–Zr system.

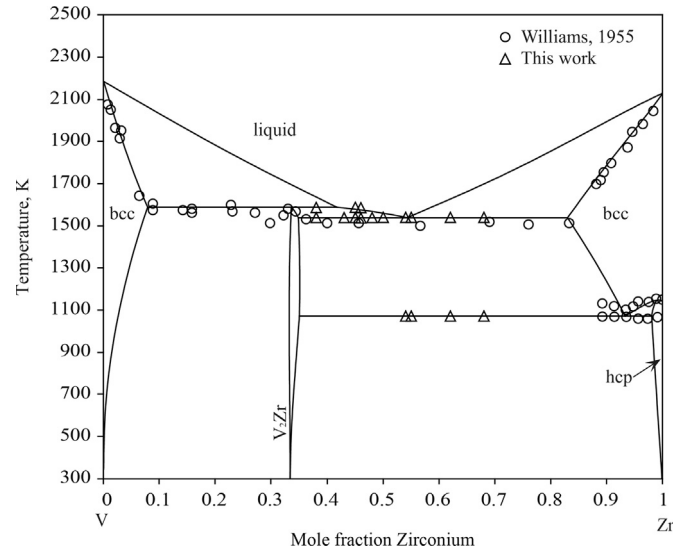
Reaction	T (K)		x(Zr)		Refs.
liquid + bcc(V) → V ₂ Zr	1586	0.4169	0.07853	0.3354	Calculated in this work
	1585	~0.42	–	–	Measured in this work
	~2013	~0.0200	–	–	Measured [10]
	1573	–	–	–	Measured [11]
	1573	~0.0500	~0.4800	0.330	Assessed [13]
	1553	0.0460	0.4941	0.3333	Calculated [18]
	1586	0.0490	0.4780	0.3350	Calculated [19]
	1573	0.0571	0.4790	0.3396	Calculated [20]
liquid → V ₂ Zr + bcc(Zr)	1537	0.5429	0.3481	0.8295	Calculated in this work
	1537	~0.54	–	–	Measured in this work
	~1633	–	–	–	Measured [10]
	1503	–	–	–	Measured [11]
	1538	0.5700	~0.3500	~0.8350	Assessed [13]
	1510	0.5477	0.3333	0.8393	Calculated [18]
	1525	0.5610	0.3520	0.8310	Calculated [19]
	1538	0.5482	0.3435	0.8313	Calculated [20]
bcc(Zr) → V ₂ Zr + hcp(Zr)	1070	0.9326	0.3500	0.9800	Calculated in this work
	1070	–	–	–	Measured in this work
	1050	–	–	–	Measured [11]
	1050	~0.9140	~0.3450	~0.99	Assessed [13]
	1063	0.9273	0.3333	0.9592	Calculated [18]
	1046	0.9220	0.3430	0.9690	Calculated [19]
	1050	0.9287	0.3496	0.9968	Calculated [20]

Table 3Thermodynamic parameters in the V–Zr system^a.

Phase	Thermodynamic parameters
liquid	Model (V, Zr) ₁
	$0_{L,V,Zr}^{\text{liq}} = -15937.91 + 12.5173T$
	$1_{L,V,Zr}^{\text{liq}} = -10361.01 + 7.1042T$
	$2_{L,V,Zr}^{\text{liq}} = -5976.59$
bcc (V)	Model (V, Zr) ₁ (Va) ₃
	$0_{L,V,Zr}^{\text{bcc}} = 17937.75 + 8.6005T$
	$1_{L,V,Zr}^{\text{bcc}} = -5335.10 + 7.3360T$
	$2_{L,V,Zr}^{\text{bcc}} = -5308.12$
hcp (Zr)	Model (V, Zr) ₁ (Va) _{0.5}
	$0_{L,V,Zr}^{\text{hcp}} = 244469.21 + 2.0608T$
	$1_{L,V,Zr}^{\text{hcp}} = 241105.53$
	$2_{L,V,Zr}^{\text{hcp}} = 241105.53$
V ₂ Zr	Model (V, Zr) ₂ (V,Zr)
	$G_{V_2Zr}^{\text{V}_2Zr} = 2\text{GHSE}_{V_2Zr} + \text{GHSE}_{Zr} - 12672.96 + 3.8143T$
	$G_{Zr_2V}^{\text{V}_2Zr} = \text{GHSE}_{V_2Zr} + 2\text{GHSE}_{Zr} + 42672.96 - 3.8143T$
	$G_{V_2V}^{\text{V}_2Zr} = 3\text{GHSE}_{V_2Zr} + 15000$
	$G_{Zr_2Zr}^{\text{V}_2Zr} = 3\text{GHSE}_{Zr} + 15000$
	$0_{L,V_2Zr}^{\text{V}_2Zr} = 11001.76 + 9.5308T$
	$0_{L,V_2Zr}^{\text{V}_2Zr} = 18017.28 + 30.7673T$

^a In J mol^{−1} of the formula units.

phase bcc(Zr) should be formed during the eutectic reaction of liquid → bcc(Zr) + V₂Zr, and hcp(Zr) was from the eutectoid reaction bcc(Zr) → hcp(Zr) + V₂Zr. So the bcc(Zr) and hcp(Zr) phases should exist in the lamellar microstructure. Similar results are observed for alloys #3–#6 as shown in Fig. 3(c–d). These results are consistent with the diffraction pattern of alloys #2–#6 as shown in Fig. 2. It is shown that V₂Zr is solidified directly from liquid phase, and the peritectic reaction, liquid + bcc(V) → V₂Zr, is not occurred in alloy #2, so it can be confirmed that the composition of the liquid in the peritectic reaction liquid + bcc(V) → V₂Zr is located between the compositions of the alloys #1 and #2.

**Fig. 8.** Calculated V–Zr phase diagram using the present thermodynamic description in comparison with the present experimental results and the experimental data [11].

As shown in Fig. 4(b), according to its hairline metallographic observations, the microstructure region of the binary eutectic microstructure is obviously present. Moreover, the uniform distribution microstructure demonstrates that the liquid composition of the eutectic reaction, liquid → V₂Zr + bcc(Zr), is close to the composition point of sample #7 which is about 54 at% Zr. The result is in agreement with the XRD diffractogram of alloy #7 as shown in Fig. 4(a), in which the diffraction peaks of hcp(Zr), bcc(Zr) and V₂Zr are observed.

Fig. 5 shows the BEI micrographs of alloy #9 (V–62 at% Zr) and alloy #10 (V–68 at% Zr). Two different microstructure regions, the bright dendrite microstructure and the lamellar microstructure, are observed in each micrograph. Fig. 5(a) is the BEI micrograph of alloy #9, the bright dendrite microstructure is the primary solidification phase with the composition of V–92.58 at% Zr by EDS, and is the bcc(Zr) phase. Similar result was observed for alloy #10.

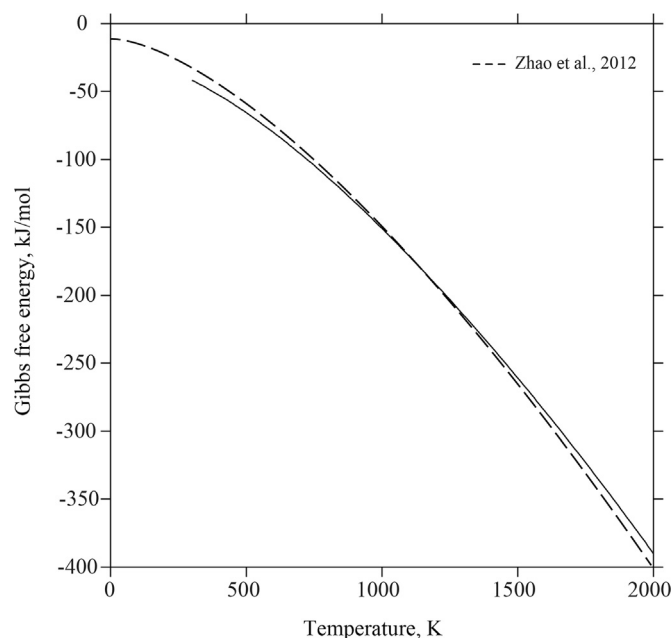


Fig. 9. Calculated Gibbs free energy of the V_2Zr and comparison with the calculated results by Zhao et al. [20].

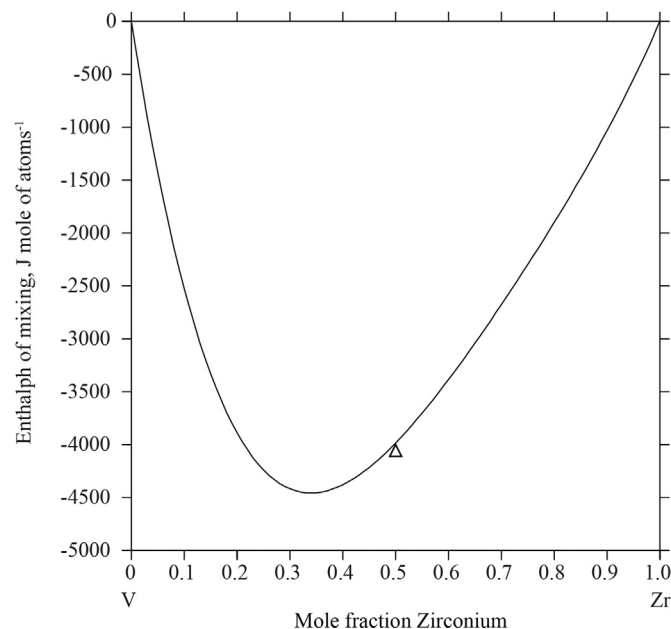


Fig. 10. Calculated enthalpies of mixing of liquid at 2400 K in the V–Zr system and comparison with the one predicted by de Boer et al. [25] using the Miedema method [26].

DTA is performed to determine the phase transformation temperature, and the typical curves of alloys #3, #4, #7 and #10 on heating are presented in Figs. 6 and 7. Fig. 6 gives the DTA heating curves of alloys #3 (V–45 at% Zr) and #4 (V–46 at% Zr) with a heating rate of 10 K/min. Both of the curves show two clearly visible peaks. According to BEI analysis on the alloys and the phase equilibria of the V–Zr system, the peaks of 1535 K and 1537 K are corresponding to the temperature of the eutectic reaction, liquid $\rightarrow V_2Zr + bcc(Zr)$, and the peaks of 1584 K and 1586 K presents the temperature of the phase transformation, liquid $\rightarrow V_2Zr$. Fig. 7 shows the DTA heating curves of alloys #7 (V–54 at% Zr) and #10

(V–68 at% Zr). There are also two peaks in each curve. It can be deduced that the first one with an onset at 1070 K represents the temperature of the eutectoid reaction, $bcc(Zr) \rightarrow V_2Zr + hcp(Zr)$, and the second one at 1538 K is that of the eutectic reaction, liquid $\rightarrow V_2Zr + bcc(Zr)$. The temperatures of phase transformation in other six samples measured by DTA are listed in Table 1.

5.2. Thermodynamic optimization

The optimization is carried out by means of the optimization module PARROT of the thermodynamic software Thermo-Calc [24], which can handle various kinds of experimental data. After all the experimental data on the phase diagram and thermodynamic properties are introduced in the program, the program works by minimizing an error of sum where each of the selected data values is given a certain weight. The weight is chosen by personal judgment and changed by trial and error during the work until most of the selected experimental information is reproduced within the expected uncertainty limits. All thermodynamic calculations are carried out using Thermo-Calc software. A set of complete and self-consistent thermodynamic description for the V–Zr system is obtained and listed in Table 3.

The optimization procedure started with the V_2Zr phase since the existence of the calculated Gibbs free energy of V_2Zr [20]. The V_2Zr phase was described as a two-sublattice model, the interaction parameters of V_2Zr in each sublattice were evaluated based on the experimental homogeneity ranges and phase relations. In the second step, the thermodynamic parameters of liquid, $bcc(Zr)$ and $hcp(Zr)$ were optimized to reproduce the experimental data of the liquidus phase, the solubility of $bcc(Zr)$ and $hcp(Zr)$, and the invariant reactions, liquid + $bcc(V) \rightarrow V_2Zr$, liquid $\rightarrow V_2Zr + bcc(Zr)$, and $bcc(Zr) \rightarrow V_2Zr + hcp(Zr)$. Due to no experimental enthalpy of mixing of the liquid, the thermodynamic parameters of liquid were optimized according to the predicted enthalpies of mixing of the equiatomic liquid phase by de Boer et al. [25] using the Miedema method [26] and the phase relations between the liquid phase and other phases. Many iterations were necessary to reproduce all experimental data. Finally, the model parameters of all phases were optimized simultaneously with all experimental data included.

Table 2 compares the calculated and measured invariant equilibria in the V–Zr system. The agreement using the present modeling is judged satisfactorily. The thermodynamic parameters of the V–Zr system obtained in the present work are listed in Table 3.

Fig. 8 presents the calculated V–Zr phase diagram using the present thermodynamic description in comparison with the present experimental results and the experimental data from Williams [11]. During constructing $bcc(Zr) + V_2Zr$ two-phase region, the phase boundary of $bcc(Zr)$ side was not completely constructed according to the experimental data in the V–Zr phase diagram measured by Williams [11]. As for the solubility of V in $hcp(Zr)$, it is optimized according to the experimental ternary Ti–V–Zr system [27]. Obviously, the calculated result is in good agreement with the corresponding experiment data in the present experimental work and the literature.

Fig. 9 is the calculated Gibbs free energy of the V_2Zr phase per mole formula unit and the comparison with the first-principles calculations by Zhao et al. [20]. It can be seen that the calculation can coincide with the calculated results from Zhao et al. [20] satisfactorily.

Fig. 10 is the calculated enthalpies of mixing of liquid at 2400 K in the V–Zr system in comparison with the one predicted by de Boer et al. [25] using the Miedema method [26]. A good agreement is obtained.

6. Conclusions

10 alloys were prepared and analyzed by means of SEM-EDS, XRD and DTA analysis to supply new phase equilibrium data for the thermodynamic modeling of the V–Zr system. The peritectic reaction, liquid + bcc(V) → V₂Zr, and the eutectic reaction, liquid → V₂Zr + bcc(Zr), were determined by SEM-EDS and XRD. The temperatures of three invariant reactions, liquid → V₂Zr + bcc(Zr), liquid + bcc(V) → V₂Zr and bcc(Zr) → V₂Zr + hcp(Zr), were measured by DTA. Considering the present experimental results and the critical assessed literature data, the thermodynamic optimization of the V–Zr system was carried out by the CALPHAD method. A set of self-consistent thermodynamic description for the V–Zr system was obtained. The comprehensive comparison shows that the calculated phase diagram and thermodynamic properties are in a good agreement with the experimental data.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.calphad.2016.03.008>.

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